

Summer 2025 Solution

2 marks

1. Attempt any FIVE of the following :

a) Enlist four basic characteristics for efficient data communication system.

Ans.

1. Delivery – Data must be delivered to the correct destination.
2. Accuracy – Data should be delivered without errors.
3. Timeliness – Data must be delivered within an acceptable time limit.
4. Low Jitter – Variation in packet arrival time should be minimal.

b) The bit rate of a signal is 2000. If each signal element carries five bits. What is the baud rate?

Ans.

- Bit rate = 2000 bps
- Bits per signal element = 5

$$\text{Baud Rate} = \frac{\text{Bit Rate}}{\text{Bits per Signal Element}} = \frac{2000}{5} = 400 \text{ baud}$$

Answer: Baud rate = 400 baud

c) State four disadvantages of line of sight Transmission (LOS) with respect to Data Communication.

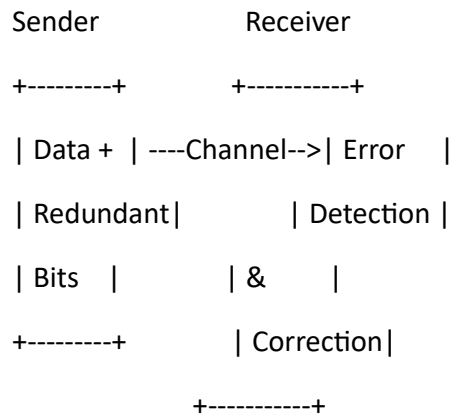
Ans.

1. Requires clear, unobstructed path between transmitter and receiver.
2. Affected by weather conditions like fog, rain, and dust.
3. Limited range due to Earth's curvature.
4. High installation and maintenance cost (towers, antennas).

d) Define Forward Error Correction (FEC) Technique with neat labelled diagram/sketch.

Ans. Forward Error Correction (FEC) is an error control technique in which the sender adds redundant bits to the data so that the receiver can detect and correct errors without retransmission.

Neat labelled sketch:



e) Draw the neat sketch with labels layered Architecture of seven layers of OSI/ISO reference model for Data communication.

Ans.

Layer No.	Layer Name	Key Function
7	Application	Provides network services directly to user applications (HTTP, FTP).
6	Presentation	Handles data formatting, encryption, and compression.
5	Session	Manages, maintains, and terminates connections between applications.
4	Transport	Ensures reliable end-to-end data transfer and flow control (TCP/UDP).
3	Network	Handles logical addressing and routing of packets (IP, ICMP).
2	Data Link	Provides physical addressing (MAC) and error-free frame transfer.

Layer No.	Layer Name	Key Function
1	Physical	Transmits raw bitstreams over physical media (cables, radio).

f) Enlist four key FTP standardized commands to facilitate communication between client and server.

Ans.

1. USER – Specifies username
2. PASS – Specifies password
3. RETR – Retrieves a file from server
4. STOR – Stores a file on server

g) State four requirements of Network computing Model to share resources and information

Ans.

1. Resource sharing (hardware, software, data)
2. Reliable communication between systems
3. Scalability to support growth
4. Security for data and resource access

4 marks

2. Attempt any THREE of the following :

a) Give difference between Analog Signal and Digital Signal with respect to terms.

i) Representation ii) Diagram Bandwidth iii) Data Storage Application iv) Application/uses

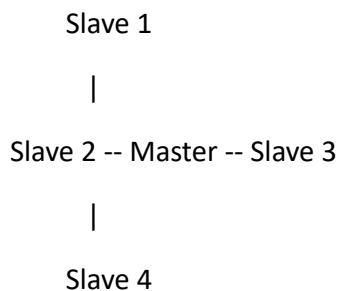
Ans.

Term	Analog Signal	Digital Signal
i) Representation	Continuous waveform with infinite values	Discrete waveform with binary values (0 and 1)
ii) Diagram / Bandwidth	Uses continuous sine wave; generally requires higher bandwidth	Uses square wave; requires less bandwidth

Term	Analog Signal	Digital Signal
iii) Data Storage Application	Difficult to store and copy without distortion	Easy to store, copy, and process without loss
iv) Application / Uses	Radio, telephone, audio transmission	Computers, internet, mobile communication

b) Draw and explain bluetooth architecture of piconet.

Ans. Neat labelled sketch (Piconet)



Explanation:

- A piconet is a basic Bluetooth network.
- It consists of one Master device and up to seven active Slave devices.
- The master controls communication, timing, and frequency hopping.
- All slaves communicate only with the master, not directly with each other.
- Used in short-range communication such as keyboards, headsets, and smartphones.

c) Explain the working of ppp (point to point) protocol.

Ans. PPP is a **data link layer protocol** used for direct communication between two nodes.

Working steps:

- 1. Link Establishment**
 - PPP establishes a link using **LCP (Link Control Protocol)**.
- 2. Authentication (Optional)**
 - Uses **PAP or CHAP** to authenticate users.
- 3. Data Transfer**
 - Data is encapsulated into PPP frames and transmitted.
- 4. Link Termination**
 - Connection is terminated after communication.

Features:

- Supports **error detection, authentication, and multiple protocols.**

d) State the functions of Internet Protocol (IP) considering below mention functions.

i) Logical Addressing ii) Packet Routing iii) Fragmentation iv) Decapsulation

Ans. i) Logical Addressing

- IP assigns logical (IP) addresses to uniquely identify source and destination hosts.

ii) Packet Routing

- IP determines the best path for packets using routing tables and routers.

iii) Fragmentation

- IP divides large packets into smaller fragments to match network MTU size.

iv) Decapsulation

- At the destination, IP removes IP header and passes data to the Transport Layer.

3. Attempt any THREE of the following :

a) Explain the terms related to Bandwidth in Data communication and networking.

i) Amplitude with diagram and units used to measure amplitude.

ii) Frequency with diagram and units used to measure frequency.

iii) Wavelength with diagram and formula used to calculate wavelength.

iv) Phase with diagram and units used to measure phase.

Ans. i) Amplitude

Amplitude refers to the maximum displacement or the "height" of a wave from its equilibrium (rest) position. In networking, it represents the strength or power of the signal. A higher amplitude generally means a stronger signal that can travel further or carry more energy.

- Units: Measured in Volts (V) for electrical signals, Watts (W) for power, or Decibels (dB) for relative signal strength.

ii) Frequency

Frequency is the number of complete cycles or oscillations a wave makes in a specific amount of time (usually one second). It represents the rate of change of the signal; a signal that changes rapidly has a high frequency, while a slow-changing signal has a low frequency.

- Units: Measured in Hertz (Hz).
 - $1\text{ Hz} = 1\text{ cycle per second}$
 - Common multiples include kHz (10^3), MHz (10^6), and GHz (10^9).

iii) Wavelength

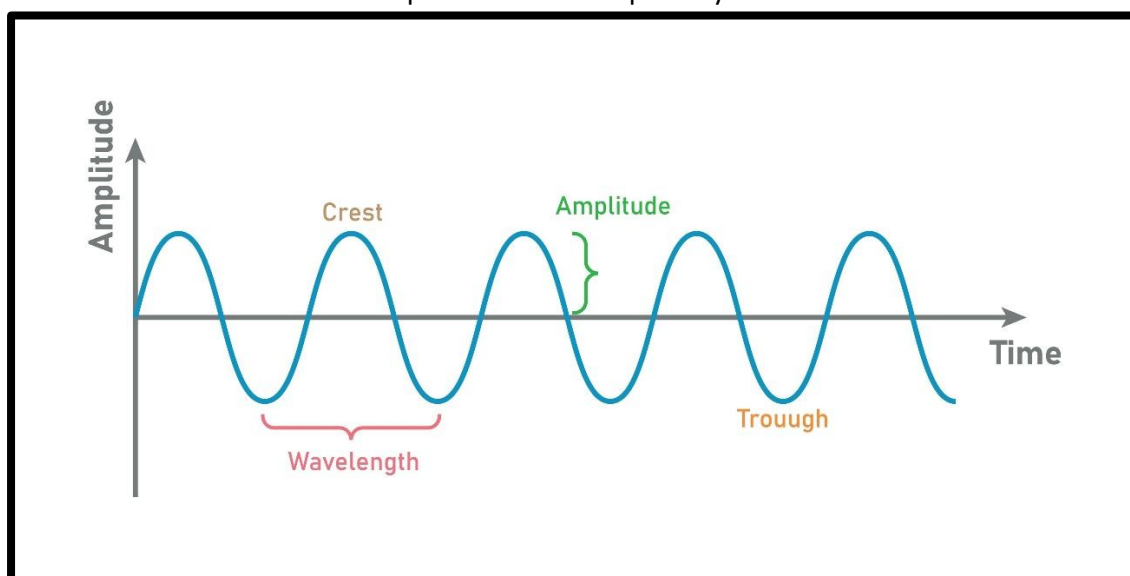
Wavelength is the physical distance between two identical points on consecutive cycles of a wave, such as the distance from one crest to the next. While frequency is measured against time, wavelength is measured against distance in space.

- Formula: $\lambda = \frac{v}{f}$
 - Where λ (Lambda) is the wavelength.
 - v is the propagation speed of the signal (e.g., speed of light $c \approx 3 \times 10^8\text{ m/s}$ in a vacuum).
 - f is the frequency.
- Units: Measured in Meters (m) or micrometers (μm) for fiber optics.

iv) Phase

Phase describes the position of the waveform relative to a reference point (usually time $t=0$). It indicates how much of the cycle has elapsed. In networking, "phase shift" is often used to modulate data, as seen in techniques like PSK (Phase Shift Keying).

- Units: Measured in Degrees (0° to 360°) or Radians (0 to 2π).
 - A shift of 360° represents one complete cycle.



b) Explain the Richard W. Hamming Error detection code for Error present in the received data and Error correction code for the Error present in the received data.

Ans. Hamming Code is an error control technique used to detect and correct single-bit errors in transmitted data.

Error Detection:

- Redundant parity bits are added at specific positions.
- Receiver recalculates parity bits.
- If parity mismatch occurs, error is detected.

Error Correction:

- Parity bits form a binary syndrome value.
- Syndrome gives the exact position of the error bit.
- Receiver corrects the error without retransmission.

Features:

- Corrects single-bit error
- Detects double-bit error
- Widely used in memory systems and digital communication

c) Explain three server methods of Dynamic Host Configuration Protocol (DHCP) of TCP/IP network model for allocating IP – addresses. State two features of DHCP (Dynamic Host Configuration Protocol).

Ans. In a TCP/IP network model, the Dynamic Host Configuration Protocol (DHCP) is an application layer protocol used to automate the assignment of IP addresses. It uses a client-server architecture where a DHCP server manages a pool of addresses and configuration parameters.

A DHCP server can be configured to assign IP addresses using these three distinct mechanisms:

1. Dynamic Allocation

This is the most common method. The server assigns an IP address from a "pool" of available addresses for a limited period, known as a lease.

- How it works: When the lease expires or the client disconnects, the IP address is returned to the pool so it can be reused by other devices.

- Use Case: Perfect for guest Wi-Fi networks or large offices where devices frequently join and leave.

2. Automatic Allocation

In this method, the DHCP server assigns a permanent IP address to a client from its available pool.

- How it works: Unlike dynamic allocation, the server keeps a record of which IP was assigned to which device. When the device reconnects in the future, the server automatically gives it the same IP address.
- Use Case: Suitable for internal workstations that do not require a specific "static" IP but benefit from having a consistent one.

3. Manual (Static) Allocation

This is also known as DHCP Reservation. The network administrator manually maps a specific IP address to a device's unique physical address (MAC address).

- How it works: When the DHCP server sees a request from a specific MAC address, it always provides the pre-assigned IP address. The address is never given to any other device.
- Use Case: Critical for servers, network printers, or routers that must be reachable at a fixed, unchanging address.

Two Features of DHCP :

1. Centralized Management: DHCP allows network administrators to manage all IP addresses and network parameters (like DNS servers and Default Gateways) from a single central server. This eliminates the need to manually configure every individual device on the network.
2. Prevention of IP Conflicts: The DHCP server tracks all assigned addresses in a database. By automating the process, it ensures that no two devices are assigned the same IP address simultaneously, preventing communication errors and network downtime.

d) Explain the following methods of switches for OSI network model

i) store – and – forward switching

ii) cut – through switching

Ans. In computer networking, switches operate primarily at the Data Link Layer (Layer 2) of the OSI model. They use different methods to receive and forward Ethernet frames between devices. The two main methods are Store-and-Forward and Cut-Through switching.

i) Store-and-Forward Switching

In this method, the switch waits until the entire frame has been received before it begins the forwarding process.

- **How it Works:** The switch copies the complete frame into its internal buffer. It then performs a Cyclic Redundancy Check (CRC) to ensure the frame is not corrupted. If the CRC matches, the switch looks up the destination MAC address and forwards the frame. If the frame is damaged, it is discarded.
- **Key Advantage:** It ensures error-free delivery because bad frames (runts or corrupted data) are filtered out before they consume bandwidth on the rest of the network.
- **Key Disadvantage:** It introduces higher latency (delay) because the switch must wait for the full frame to arrive and then process the checksum.

ii) Cut-Through Switching

In this method, the switch starts forwarding the frame as soon as the destination address is identified, without waiting for the rest of the data.

- **How it Works:** The switch only reads the first 6 bytes of the frame (which contains the destination MAC address). Once it knows where the frame is going, it immediately starts "streaming" the data to the outgoing port while the rest of the frame is still arriving at the input port.
- **Key Advantage:** It offers extremely low latency, making it ideal for high-performance computing or real-time applications where every microsecond counts.
- **Key Disadvantage:** It has no error checking. Because the switch doesn't wait for the Frame Check Sequence (FCS) at the end of the frame, it will unknowingly forward corrupted or incomplete frames, relying on the destination device to detect and drop them.

4. Attempt any THREE of the following :

a) Explain Full Duplex Communication Mode with neat labelled diagram for Data Communication. State two advantages of Full Duplex Communication. State two disadvantages of Full Duplex Communication.

Ans. 1. Definition

Full-Duplex mode is a bidirectional communication mode where the communication link is shared by both signals traveling in opposite directions simultaneously. This is typically achieved by having two separate transmission paths (one for sending and one for receiving) or by dividing the capacity of a single channel into two separate "lanes."

Example: A telephone conversation is a classic example of full-duplex communication; both parties can speak and hear each other at the exact same time.

2. Labelled Diagram of Full-Duplex Mode

The diagram below illustrates how two devices communicate simultaneously over a link.

3. Advantages of Full-Duplex Communication

- **Maximum Performance and Speed:** Since data can be sent and received at the same time, the throughput is doubled compared to half-duplex systems. There is no waiting time for the channel to become free.
- **Collision-Free Environment:** In networking (like Full-Duplex Ethernet), sending and receiving use separate "lanes" or wires. This eliminates data collisions, which significantly improves network efficiency and reduces the need for retransmissions.

4. Disadvantages of Full-Duplex Communication

- **High Implementation Cost:** Full-duplex systems require more sophisticated hardware (such as specialized switches and NICs) and more complex cabling (like two-pair twisted cables or dual fiber strands) to handle simultaneous signals.
- **Increased Bandwidth Consumption:** Because the system maintains two active paths for communication at all times, it utilizes more of the available bandwidth compared to simplex or half-duplex modes, which can be a limitation in spectrum-constrained environments like some wireless radios.

b) Explain the working of satellite communication system in detail with signal uplink and signal downlink along with neat labelled diagram.

Ans. In data communication, a Satellite Communication System acts as a microwave relay station in space. It is used to transfer information (voice, video, or data) between two or more points on Earth that are separated by large distances or obstacles like oceans and mountains.

The system is broadly divided into two segments: the Ground Segment (Earth Stations) and the Space Segment (The Satellite).

Working of Satellite Communication

The working involves three primary stages: Uplink, Transponder Processing, and Downlink.

1. Signal Uplink

The process begins at a Transmitting Earth Station. The data or signal is modulated onto a high-frequency carrier wave.

- Definition: Uplink is the transmission of a signal from an Earth station to the satellite.
- Frequency: High frequencies are used (e.g., 6 GHz for C-band) to penetrate the Earth's atmosphere effectively.
- Process: A high-power amplifier (HPA) boosts the signal, and a large parabolic dish antenna beams it precisely toward the satellite's orbital position.

2. The Satellite Transponder (The Relay)

Once the satellite receives the weak uplink signal, its onboard electronic system, called a Transponder, takes over.

- Amplification: The signal, weakened by the long travel through space, is amplified using a Low Noise Amplifier (LNA).
- Frequency Translation: The satellite changes the frequency of the signal. This is crucial to prevent interference between the powerful outgoing signal and the weak incoming signal.
- Re-transmission: The signal is then boosted again by a Power Amplifier (like a TWTA) for its journey back to Earth.

3. Signal Downlink

The processed signal is then beamed back toward Earth to be picked up by one or more receiving stations.

- Definition: Downlink is the transmission of a signal from the satellite back to an Earth station.
- Frequency: This frequency is always lower than the uplink frequency (e.g., 4 GHz for C-band) to minimize atmospheric interference and power consumption on the satellite.
- Reception: The Receiving Earth Station uses its dish antenna to capture the signal, which is then demodulated to recover the original data.

c) Explain two types of Error in Data transmission with example.

i) Single – bit Error

ii) Burst Error

Ans. In data communication, errors occur when the signal is altered during transmission due to interference or noise. These errors are generally classified into two types based on the number of bits affected.

i) Single-Bit Error

A Single-bit error occurs when only one bit of a given data unit (such as a byte or packet) is changed from 1 to 0 or from 0 to 1.

- Occurrence: These are the least likely errors in serial data transmission but can occur frequently in parallel transmission (where multiple wires carry bits simultaneously and one wire is noisy).
- Example: * Sent Data: 01011101
 - Received Data: 0101**0**101
 - Result: Only the 5th bit was flipped from 1 to 0.

ii) Burst Error

A Burst error occurs when two or more bits in a data unit have changed. A burst error does not necessarily mean that the corrupted bits are consecutive.

- Length of Burst: The length is measured from the first corrupted bit to the last corrupted bit in the sequence. Some bits in between may remain uncorrupted.
- Occurrence: Burst errors are very common in serial transmission. Since noise (like lightning or electromagnetic interference) usually lasts longer than the duration of a single bit, it typically affects a group of bits.

- Example:
 - Sent Data: 0100010001000011
 - Received Data: 01**011101011**00011
 - *Result:* Multiple bits between the 4th and 11th positions are corrupted. The length of this burst error is 8 bits.

d) Explain the structure of IPV6 Address in detail.

Ans. An Internet Protocol version 6 (IPv6) address is a 128-bit alphanumeric identifier used to identify a device on a network. It was developed by the IETF to replace IPv4, which only uses 32-bit addresses and has run out of available space.

1. General Format

An IPv6 address is 128 bits long, which is four times the size of an IPv4 address. It is represented as eight groups of four hexadecimal digits, each group representing 16 bits. These groups are separated by colons (:).

- Total Bits: 128 bits
- Notation: Hexadecimal (0-9 and A-F)
- Groups: 8 groups (called "hextets")
- Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

2. Hierarchical Structure

The 128 bits of an IPv6 address are typically divided into two main parts, each 64 bits long:

A. Network Portion (First 64 bits)

This part is used for routing and identifies the specific network to which the device belongs. It is further subdivided into:

1. Global Routing Prefix (48 bits): Assigned by Internet Service Providers (ISPs) to a customer or site. It represents the public topology.
2. Subnet ID (16 bits): Used by an organization to identify subnets within its site. This allows for 2^{16} (65,536) unique subnets.

B. Interface ID / Host Portion (Last 64 bits)

This part uniquely identifies a specific interface (device) on the subnet. It is often automatically generated using the device's MAC address (via the EUI-64 process) or assigned randomly for privacy.

3. Rules for Address Compression

Since IPv6 addresses are very long, two rules are used to shorten them:

- Rule 1: Omit Leading Zeros

Within any 16-bit group, leading zeros can be ignored.

- *Example:* 0db8 becomes db8, and 0000 becomes 0.

- Rule 2: Double Colon (::) Substitution

A single contiguous sequence of all-zero groups can be replaced with a double colon ::.

- Note: This can only be used once per address to avoid ambiguity.
- *Example:* 2001:db8:0000:0000:0000:0000:0000:1 becomes 2001:db8::1.

e) Explain two functions of following layer of OSI/ISO Model used for i) Session layer ii)

Presentation layer

Ans. In the OSI (Open Systems Interconnection) model, both the Session and Presentation layers handle high-level data management and formatting before data is transmitted over the network.

i) Session Layer (Layer 5)

The Session Layer is responsible for establishing, managing, and terminating "sessions" or dialogues between applications on different devices.

Two Key Functions:

1. Dialogue Control: It acts as a traffic controller, determining how the conversation flows. It can establish communication in Half-Duplex (one way at a time) or Full-Duplex (both ways simultaneously). It manages "tokens" to ensure that devices don't try to perform critical operations at the same time.
2. Synchronization (Checkpointing): It allows a process to add checkpoints (synchronization points) into a data stream.
 - *Example:* If you are downloading a large 100MB file and the connection fails at 65MB, the Session Layer can use a checkpoint at 60MB to resume the transfer from that point rather than starting the entire download from scratch.

ii) Presentation Layer (Layer 6)

The Presentation Layer acts as a translator. It ensures that the information sent by the Application Layer of one system is readable by the Application Layer of another.

Two Key Functions:

1. **Data Translation:** Different computers use different encoding systems (e.g., one might use ASCII and another EBCDIC). This layer translates the data into a common bitstream format that both systems can understand.
2. **Encryption and Decryption:** To ensure security during transmission, the Presentation Layer can encrypt data (convert it to a hidden code) at the sender's side and decrypt it (convert it back to readable text) at the receiver's side. It is also responsible for Data Compression, which reduces the number of bits to be transmitted, making the communication more efficient.

6 marks

5. Attempt any TWO of the following :

a) Explain the following types of network with neat label diagram.

i) LAN ii) MAN iii) WAN

Ans. Computer networks are categorized primarily based on their geographical size, the number of users they support, and the speed of data transmission.

i) LAN (Local Area Network)

A LAN is a private network that connects computers and devices within a limited geographical area, such as a single room, a home, an office building, or a school campus.

- **Geographical Area:** Small (usually up to 1–2 km).
- **Speed:** Very high data transfer rates (100 Mbps to 10 Gbps).
- **Ownership:** Privately owned by an individual or a single organization.
- **Media:** Typically uses Ethernet cables (Twisted-pair) or Wi-Fi (WLAN).

ii) MAN (Metropolitan Area Network)

A MAN is a larger network that interconnects several LANs across a city or a large campus. It acts as a bridge between the local speed of a LAN and the vast reach of a WAN.

- **Geographical Area:** Medium (typically 10–50 km radius).
- **Speed:** Moderate to high (usually 155 Mbps to 1 Gbps).
- **Ownership:** Can be owned by a single large organization (like a university) or a public service provider (like a cable TV company).
- **Example:** A city-wide cable TV network or a university campus connecting multiple departmental buildings.

iii) WAN (Wide Area Network)

A WAN is a network that covers a very large geographical area, such as a state, a country, or even the entire world. It often connects smaller networks (LANs and MANs) together.

- Geographical Area: Vast (spanning countries or continents).
- Speed: Lower than LANs due to the long distances involved (typically 1 Mbps to 100 Mbps).
- Ownership: Not usually owned by a single organization; it is maintained by multiple ISPs and telecommunication companies.
- Media: Uses satellite links, fiber-optic undersea cables, and leased telephone lines.
- Example: The Internet is the most well-known example of a public WAN.

b) Explain three phases of connection oriented packet switching with neat labelled sketch.

State two advantages of packet switching.

State two disadvantages of packet switching.

Ans. In data communication, Connection-Oriented Packet Switching (also known as Virtual Circuit Switching) mimics the behavior of circuit switching by establishing a logical path before any data is sent. This ensures that all packets belonging to a single message follow the same route and arrive in the correct order.

The process follows a strict sequence of three phases:

1. **Connection Setup Phase:** Before data transmission begins, the source sends a special Request Packet to the destination. This packet travels through the network, and each intermediate switch creates a table entry (mapping an incoming port and Virtual Circuit Identifier to an outgoing port). The destination sends back an Acknowledgment Packet to confirm the path is ready.
2. **Data Transfer Phase:** Once the virtual circuit is established, the actual data packets are sent. Since the route is already defined in the switch tables, these packets do not need to carry full destination addresses; they only carry a short Virtual Circuit Identifier (VCI). Every packet follows the exact same pre-defined path.
3. **Connection Teardown Phase:** After all data has been transmitted, the source sends a Teardown Packet. As this packet passes through the switches, they delete the

corresponding entries from their routing tables, releasing the logical resources for other connections.

Neat Labelled Sketch

Advantages:

- **High Efficiency (Bandwidth Sharing):** Unlike circuit switching, there are no dedicated lines. Multiple users can share the same physical link simultaneously through statistical multiplexing, ensuring that the network is rarely idle.
- **Reliability and Robustness:** In the event of a node or link failure, packets can be dynamically rerouted through alternative paths (in datagram mode), ensuring that the data eventually reaches its destination.

Disadvantages:

- **Variable Delay (Latency):** Because packets may be buffered (Store-and-Forward) or experience queuing at busy switches, the arrival time can vary. This makes it less ideal for real-time applications like voice calls without specialized Quality of Service (QoS).
- **Complexity and Overhead:** Every packet must contain header information (such as sequence numbers and addresses). Additionally, switching nodes require significant processing power and memory to manage packet headers and reassemble data at the destination.

c) Explain the working of TCP and UDP protocol in the TCP/IP network model with neat labelled diagram. State any two advantages of TCP and UDP protocol.

Ans. In the TCP/IP model, both TCP and UDP reside in the Transport Layer. While they both carry data, they operate with completely different philosophies: one prioritizes reliability (TCP), while the other prioritizes speed (UDP).

1. Transmission Control Protocol (TCP)

TCP is a connection-oriented protocol. It ensures that all data sent is received correctly and in the right order.

How it works:

- **Three-Way Handshake:** Before sending data, TCP establishes a connection using a three-step process: SYN (Synchronize), SYN-ACK (Synchronize-Acknowledge), and ACK (Acknowledge).

- Sequencing & Acknowledgments: Each data segment is given a sequence number. The receiver sends an "ACK" for every segment it gets.
- Retransmission: If the sender doesn't receive an ACK within a certain time, it assumes the packet was lost and sends it again.
- Flow Control: TCP regulates the speed of data transfer to ensure the receiver isn't overwhelmed.

Two Advantages of TCP:

1. High Reliability: Guarantees that data reaches the destination without errors and in the original sequence.
2. Error Correction: Automatically detects corrupted or lost packets and retransmits them.

2. User Datagram Protocol (UDP)

UDP is a connectionless protocol. It is often called a "fire-and-forget" protocol because it sends data without checking if the receiver is ready or if the data actually arrived.

How it works:

- No Handshake: UDP starts sending data immediately without any prior communication or setup.
- Independent Datagrams: Each packet (datagram) is treated as an independent unit. They may arrive out of order or not at all.
- Low Overhead: Unlike TCP, which has a 20-byte header, UDP has a tiny 8-byte header, making it much faster to process.

Two Advantages of UDP:

1. Low Latency (High Speed): Without handshakes or retransmissions, data moves much faster, which is critical for real-time apps.
2. Efficiency: It consumes fewer network resources and less CPU power because it doesn't need to track connections or sequence numbers.

6. Attempt any TWO of the following :

a) Define Time Division Multiplexing (TDM) Explain two types of Time Division

Multiplexing with neat sketch. State two applications of TDM.

Ans. In data communication and networking, Time Division Multiplexing (TDM) is a digital technique used to transmit multiple signals over a single communication channel by dividing the total available time into several non-overlapping slots.

Definition

Time Division Multiplexing (TDM) is a process where multiple low-speed digital signals are combined into one high-speed signal. Instead of sharing the frequency (like in FDM), each signal is given the entire bandwidth of the channel for a very brief, fixed duration of time (a time slot).

Two Types of Time Division Multiplexing

1. Synchronous TDM

In Synchronous TDM, the multiplexer assigns a fixed time slot to every input device in a round-robin fashion, regardless of whether the device has data to send.

- **Working:** If a device is idle, its assigned time slot remains empty in the output frame. This can lead to a waste of bandwidth.
- **Feature:** The timing is predictable and the implementation is simple.

2. Statistical (Asynchronous) TDM

Statistical TDM is a more efficient version where time slots are allocated dynamically only to those input devices that actually have data to transmit.

- **Working:** The multiplexer scans the input lines and skips any that are idle. Because the slots are not fixed, each slot must carry an address to identify which device the data belongs to.
- **Feature:** It maximizes bandwidth utilization but requires more complex hardware and introduces slight delays due to addressing.

Two Applications of TDM

1. **Digital Telephony:** TDM is used in the T-1 and E-1 carrier systems. For example, a T-1 line uses TDM to multiplex 24 separate voice calls into a single high-speed digital signal.
2. **Cellular Networks (GSM):** In 2G mobile communication (GSM), TDM allows multiple users to share the same frequency channel by assigning each user a specific time slot to transmit their voice or data.

b) Explain three types of ARQ (Automatic Repeat Request) error correction technique for data transmission with neat sketch.

Ans. Automatic Repeat Request (ARQ) is an error-control strategy used in data link and transport layers. It relies on acknowledgments (ACKs) and timeouts to ensure that data is transmitted reliably over an imperfect communication channel.

Here are the three types of ARQ techniques:

i) Stop-and-Wait ARQ

This is the simplest form of ARQ. The sender sends one frame and then stops to wait for an acknowledgment from the receiver before sending the next one.

- **Working:** If the ACK arrives within a set time (timeout), the next frame is sent. If the frame or ACK is lost, the timer expires, and the sender retransmits the same frame.
- **Window Size:** Both Sender and Receiver windows are 1.
- **Efficiency:** Very low, especially over long distances, because the sender sits idle while waiting for the ACK.

ii) Go-Back-N (GBN) ARQ

To improve efficiency, Go-Back-N allows the sender to send multiple frames (a "window" of frames) without waiting for an acknowledgment.

- **Working:** The receiver only accepts frames in order. If one frame in the middle is lost or corrupted, the receiver discards all subsequent frames (even if they are correct). The sender's timer for the lost frame will eventually expire, and the sender must "go back" and retransmit the lost frame plus all subsequent frames in the window.
- **Window Size:** Sender window is N; Receiver window is 1.
- **Efficiency:** Much higher than Stop-and-Wait, but wasteful if the error rate is high.

iii) Selective Repeat ARQ

This is the most advanced and efficient ARQ technique. It improves upon Go-Back-N by only retransmitting the frames that were actually lost or damaged.

- **Working:** The receiver has a buffer to store out-of-order frames. When a frame is missing, the receiver sends a Negative Acknowledgment (NACK) or simply waits for a timeout. The sender then retransmits only the specific missing frame, not the entire window.

- Window Size: Both Sender and Receiver windows are N (typically 2^{n-1}).
- Efficiency: Highest among the three, as it minimizes unnecessary retransmissions and saves bandwidth.

c) State difference between Peer – to – peer computing model and client – server computing model considering below functions (any six)

**i) Type of computing model ii) Type of service iii) Resource sharing iv) Security
v) Performance vi) Reliable vii) Communication type**

Ans.

Function	Peer-to-Peer (P2P) Model	Client-Server Model
i) Type of Model	Decentralized: Every node has equal status and can act as both a client and a server.	Centralized: There is a clear distinction between the central server and the peripheral clients.
ii) Type of Service	Each node can request and provide services simultaneously.	The server provides services, while the client only requests them.
iii) Resource Sharing	Resources (files, printers) are distributed across all peers in the network.	Resources are centrally stored on the server and shared with clients.
iv) Security	Lower Security: Managed individually by each user on their own machine.	Higher Security: Managed centrally by the server with strict authentication and firewalls.
v) Performance	Performance decreases as more peers join (due	Performance remains stable or increases with server

Function	Peer-to-Peer (P2P) Model	Client-Server Model
	to shared bandwidth and CPU).	upgrades; the server handles the "heavy lifting."
vi) Reliability	High: No single point of failure; if one node goes down, the network still functions.	Low: If the central server fails, the entire network or service goes down.
vii) Communication Type	Direct: Peers communicate with each other directly without a middleman.	Indirect: All communication and data requests must pass through the central server.